When Does Quantum Error Mitigation Pay Off Under a Fixed Shot Budget? Accounting, Decision Reliability, and Operational Boundaries in Simulation

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Abstract

Quantum error mitigation (QEM) removes bias at the price of extra circuit executions and amplified sampling variance. We ask when that trade pays off under one declared shot budget that charges every calibration circuit, folded execution, pilot, and diagnostic, and we judge methods by the decisions their estimates support rather than by root-mean-square error (RMSE) alone. In noisy classical simulation of unmitigated (raw) estimation, readout correction, zero-noise extrapolation (ZNE), and their combination, we find three things. First, no method dominates: raw estimation is the cost-aware choice in about half of a 144-cell grid, a count that falls as the practical-equivalence threshold grows, while readout and ZNE win in disjoint noise ranges. Second, lower RMSE does not imply better decisions. In QAOA candidate selection, ZNE lowers energy RMSE yet selects worse, an inversion confirmed on new repeats (selection error 0.271 for raw against 0.379 for ZNE, $p = 0.001$). We show that, for a normally distributed decisive gap, a debiasing method improves the decision only if noise has shrunk the raw gap below $1/\sqrt{\gamma}$ of its ideal value, where γ is the method's variance amplification (0.48 for linear ZNE), and that this preference does not depend on the budget. Third, every runtime tool we tested, from validity gates and adaptive selectors to sequential and two-layer certification, controls sampling error but not differential bias, and a mismodeled readout correction does worse than none. A hardware readiness review ends in a documented no-go. All results are simulated.

1 Introduction

QEM can reduce expectation-value bias without logical encoding, but its additional circuits and statistical amplification consume finite resources [28, 18, 5]. Comparing methods at unequal cost can manufacture an apparent advantage, and even at equal cost the best method changes with circuit, observable, noise, and sampling regime. A second gap is less often measured: a mitigated estimate is usually an input to a decision, such as which of several candidates is best, and the error that matters for that decision is not the error that RMSE measures. We call unmitigated estimation *raw* throughout.

Two lines of work frame this gap. Theory shows that the sampling cost of mitigation grows exponentially with circuit size and depth for broad classes of methods [26, 27, 24]; these are bounds on estimating an expectation value. Experiments and benchmarks show that mitigated values can be accurate at useful scales [14, 15, 29] and compare techniques across circuits and devices [4, 6, 17], while recent audits warn that a reported ZNE improvement can be artefactual [16]. Neither line asks what the estimate is for. When it feeds a choice among alternatives, the relevant theory is that of ranking and selection [2], where what matters is the error in the differences between alternatives, not in their values.

We therefore ask four questions under one resource ledger. When does mitigation improve estimation at a fixed total budget? When does better estimation produce a better decision? Can runtime-available evidence tell when mitigation, an extrapolation, or a certificate should be trusted? And where do readout correction, calibration reuse, and adaptive selection stop working?

Contributions.

1. *Fixed-budget benchmarking* (Section 5). Calibration, folding, pilots, repeats, and diagnostics are charged to one budget. We report operating regimes, signal-strength strata on a frozen independent circuit set, and the total shots each method needs to reach a decision-error target.
2. *Estimator quality versus decision reliability* (Section 6). We give a common-mode/contrast decomposition of candidate errors and a condition, Proposition 1, for when a debiasing method improves a decision at a fixed budget. We use them to explain, and then confirm on new repeats, why lower RMSE can coexist with worse selection.
3. *Validity and operational boundaries* (Section 7). We evaluate ZNE validity gates, method selectors, sequential and two-layer certification, readout correction up to width 10, mismatch dose-response, and recalibra-

tion policies under drift. Each negative result is classified by mechanism (Section 8).

The contribution is a reproducible simulator benchmark with repeat-level data, explicit execution ledgers, frozen decision rules, and audited provenance (Section 9). It is not a new QEM method and not a claim about hardware performance.

2 Related Work

Table 1 places this study among the closest lines of work by what each measures.

Limits of mitigation. Takagi et al. bound the sampling cost of any mitigation strategy [26] and give universal lower bounds on it [27]; Quek et al. tighten these bounds to exponential growth in depth [24]. The variance amplification of linear ZNE used below is an elementary instance of this cost. Our results do not test these bounds; they show what the cost does to a decision at the small sizes where mitigation is affordable.

Methods and practice. ZNE was introduced with probabilistic error cancellation [28, 18] and reviewed in [9, 5]. Digital folding makes it hardware agnostic [11], and it was central to recent large-scale experiments [14, 15]. Probabilistic error cancellation with learned sparse noise models [29], Clifford data regression [7], and randomized compiling [31] have different cost structures and are not studied here. Readout correction ranges from full detector tomography [19, 3] to scalable reduced models [21].

Benchmarks and audits. Bultrini et al. unify and compare state-of-the-art techniques [4]. Qermit benchmarks mitigation volumetrically on superconducting hardware against classical simulation [6], and Mitiq provides common implementations [17]. Köster and Mauerer show that a ZNE improvement can be an artefact of the benchmark [16], which motivates our validity gates. A concurrent preprint also relates accuracy, decision risk, and sampling cost in mitigation [25].

Statistical decisions. Choosing the best of several noisy alternatives is the ranking-and-selection problem [2]. Sequential stopping goes back to Wald [30]; confidence sequences give intervals valid at every look [13] and are an alternative to the Bonferroni-split looks used here.

3 Background and Notation

Estimators and error. For repeat r, $e_r = \widehat{\mu}_r - \mu_{\text{ref}}$. We report bias, sample variance, and

$$\text{RMSE} = \sqrt{R^{-1} \sum_{r=1}^{R} e_r^2}. \tag{1}$$

Paired RMSE-gain intervals use deterministic bias-corrected and accelerated bootstrap resampling [8]. Decision-error probability is the fraction of repeats whose decision disagrees with the exact decision; Wilson intervals [32] quantify its binomial uncertainty. Paired decision comparisons use the exact sign test on discordant pairs [20], Agresti–Min intervals for paired proportions [1], and Holm's step-down correction across a family [12]. Where conditions share a circuit, intervals resample circuits as clusters.

Table 1: What each line of work measures. The columns describe the main question of each line, not every analysis in every paper.

Line of work	Quantity studied	Question
Limits [26, 27, 24]	Sampling cost of an estimate	How fast must cost grow?
Experiments [14, 15]	Mitigated values at scale	Is the value accurate?
Benchmarks [4, 6]	Accuracy across techniques	Which technique is closer?
Audits [16]	Reported improvements	Is the gain genuine?
This work	Decisions at one total budget	Does the gain change the decision?

Methods. Raw estimation spends the full budget on the target. Readout correction allocates 20% of the budget to calibration and 80% to the target; the full-matrix variant calibrates all 2^w basis states and reports ill-conditioning explicitly rather than inverting an unstable matrix [3], while tensor-product, pairwise-local, observed-subspace, and matrix-free variants use reduced calibration models [21]. Linear ZNE splits the budget over global folds $\lambda \in \{1, 3, 5\}$ and fits

$$\widehat{\mu}(\lambda) = \beta_0 + \beta_1 \lambda, \qquad \widehat{\mu}_{\text{ZNE}} = \beta_0, \tag{2}$$

without clipping out-of-range extrapolates [11]. With equal shots per fold, the intercept weights are $(13, 4, -5)/12$, so the ZNE variance is $3\sum_i w_i^2 = 4.375$ times the raw variance at the same total budget, a standard deviation factor of 2.09. Combined mitigation corrects readout at each fold and then extrapolates.

Common-mode and contrast error. For candidates $k = 1, \ldots, K$ with estimates $\widehat{E}_k = E_k + b + d_k$, where b is the error shared by all candidates and $\sum_k d_k = 0$,

$$\frac{1}{K} \sum_k (\widehat{E}_k - E_k)^2 = b^2 + \frac{1}{K} \sum_k d_k^2. \tag{3}$$

RMSE counts both terms, but the selected candidate $\arg\max_k \widehat{E}_k$ depends only on the contrasts $E_k + d_k$. A method can therefore lower RMSE by shrinking b or the errors of non-contending candidates while leaving the selection unchanged, and can worsen the selection by inflating the variance of the contrast between the leading candidates.

Decision error under bias. For two candidates with exact gap $\Delta > 0$, an estimator whose gap converges to $\Delta + \beta$ with standard deviation $\sigma/\sqrt{N}$ has, in the nor-

mal approximation,

$$P_{\mathrm{err}}(N) \approx \Phi\left(-\frac{(\Delta+\beta)\sqrt{N}}{\sigma}\right). \tag{4}$$

If the differential bias satisfies $\beta \le -\Delta$, the error rises toward one as N grows: more shots make the wrong answer more certain. A confidence interval controls only the σ term, which is the root of several negative results below.

When does debiasing help a decision? Write the raw gap's mean as $\Delta_{\mathrm{r}} = \Delta + \beta_{\mathrm{r}}$ and a mitigated gap's mean as $\Delta_{\mathrm{m}} = \Delta + \beta_{\mathrm{m}}$, and let the mitigated gap's variance be γ times the raw variance at the same total budget.

Proposition 1 *Under the normal approximation of Eq. (4), with $\Delta_{\mathrm{r}} > 0$, the mitigated decision has lower error than the raw decision at some budget N if and only if it does at every budget, and this holds if and only if*

$$\Delta_{\mathrm{m}} > \sqrt{\gamma}\,\Delta_{\mathrm{r}}. \tag{5}$$

In particular, a method that removes the gap's bias completely ($\Delta_{\mathrm{m}} = \Delta$) helps if and only if the raw attenuation $a = \Delta_{\mathrm{r}}/\Delta$ satisfies $a < 1/\sqrt{\gamma}$. If instead $\Delta_{\mathrm{r}} \le 0 < \Delta_{\mathrm{m}}$, mitigation helps at every budget.

Proof. The error probabilities are $\Phi(-\Delta_{\mathrm{r}}\sqrt{N}/\sigma)$ and $\Phi(-\Delta_{\mathrm{m}}\sqrt{N}/(\sqrt{\gamma}\,\sigma))$. Since Φ is increasing, the second is smaller exactly when $\Delta_{\mathrm{m}}/\sqrt{\gamma} > \Delta_{\mathrm{r}}$, and the factor $\sqrt{N}/\sigma > 0$ cancels, so the comparison is the same at every N. Setting $\Delta_{\mathrm{m}} = \Delta$ gives the attenuation form. If $\Delta_{\mathrm{r}} \le 0$, the raw error is at least $1/2$ while the mitigated error is below $1/2$. □

For linear ZNE with equal shots per fold, $\gamma = 4.375$, so complete debiasing pays off only when noise has shrunk the decisive gap to less than $1/\sqrt{4.375} = 0.48$ of its ideal value. Readout correction has $\gamma \ge 1/0.8$ from its calibration share, plus a factor from inverting the assignment matrix that is small when assignment errors are small, so it can pay off for much milder attenuation, but only if the readout bias differs between the candidates; a bias common to both leaves $\Delta_{\mathrm{r}} = \Delta$ and cannot help. With K candidates, the same comparison applies to the leading contender's gap, and a union bound over the other gaps gives an upper bound on the error. The proposition is a statement about the normal approximation and treats γ and the gap means as known; Section 6 tests it on data that were collected before it was written.

4 Benchmark Design

Main grid. The GHZ family prepares $|\mathrm{GHZ}_w\rangle = (|0\rangle^{\otimes w} + |1\rangle^{\otimes w})/\sqrt{2}$, with analytic observables $X^{\otimes w} = 1$ and neighboring $ZZ = 1$. The hardware-efficient family alternates fixed-seed single-qubit rotations and nearest-neighbor entanglers at depths 1, 3, and 6. The main grid crosses widths $2, 4, 6$, hardware-efficient depths $1, 3, 6$, noise levels zero/low/medium/high, shot budgets 1,024/4,096/16,384, three methods, and 50 repeats. GHZ depths 3 and 6 remain explicit `not_applicable` cells. All seeds derive from root seed 20261102. Qiskit Aer supplies ideal statevectors and finite-shot noisy simulation [23]; the main model combines independent assignment error with one- and two-qubit depolarizing channels. A separate 4,096-shot shift set tests intermediate mixed noise, asymmetric readout, coherent over-rotation, and correlated ZZ error. These are controlled synthetic stress tests, not calibrated device models, following the broader need for common QEM benchmarks [4].

Table 2: Studies, their scale, and the evidence for rule freezing. "Commit" means the rules and manifest were committed and pushed before results existed; commit order shows publication order only.

Study	Scale	Rules frozen
Main grid	144 cells, 50 repeats	With results
Combined QEM	6 cells, 2 budgets	Commit
QAOA selection	3 graphs, 5 candidates	Commit
Selector test (v2)	160 fresh cells	Commit
Signal strata (v2)	126 circuits, 7 noises	Commit
ZNE gate (v2)	256 candidates, holdout	Commit
Confirmatory	34 + 72 comparisons	Commit
Shots to reliability	40 cells, 4 methods	Commit
Selector (v3), certificate	72 tasks, 4 families	Commit
Readout boundaries	widths 4–10	Commit

Later studies. Each later study was frozen in the repository before its noisy run, as listed in Table 2. The signal, gate, selector, certification, and readout-boundary studies use an independent NumPy density-matrix simulator that agrees with Aer to 10^{-10} on depolarizing, amplitude-damping, and dephasing channels, and samples shots from exact outcome probabilities.

Fixed-budget accounting. Every method spends the same total shots. Calibration, folded executions, pilot samples, garbage folding used by validity checks, and check circuits used by drift triggers are charged to it. At width 2 and 16,384 shots, combined mitigation uses 7 circuit executions compared with 1 for raw, 3 for ZNE, and 5 for readout. Failed corrections are kept as outcomes with a named reason and count as wrong decisions; they are never dropped.

Labels and holdouts. For each completed cell, a cost-aware oracle combines RMSE with an overhead penalty; differences below the documented RMSE threshold 0.005 are `practical_tie`, and incomplete comparisons are `inconclusive`. The first implementation of the oracle could not assign the label `raw`: a raw winner has zero gain over itself, so it always reached the practical-tie branch. This paper reports corrected labels; the original labels remain in the repository. Later studies split circuits or tasks into training, validation, and holdout sets by hash rank, choose every rule on training and validation, and read the holdout once.

Table 3: Corrected cost-aware labels on the main grid. No method wins everywhere.

Cost-aware label	Cells
Practical tie	26
Raw	77
Readout	18
Linear ZNE	14
Inconclusive	9

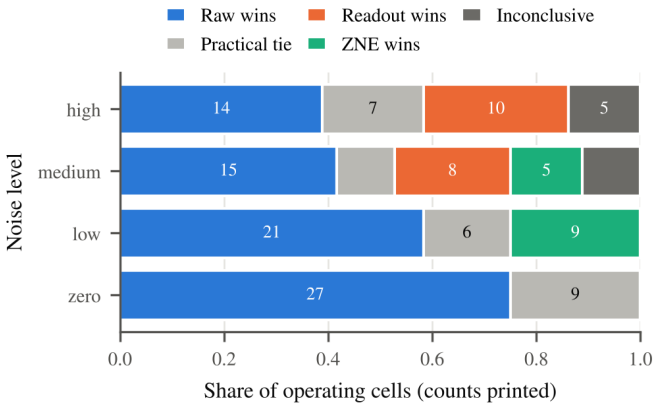


Figure 1: Cost-aware labels by noise level on the main grid at the frozen threshold. Readout and ZNE win in disjoint noise ranges; raw wins most of the remaining cells, mostly by small margins (Table 4).

5 Fixed-Budget Benchmarking

Takeaway: once every overhead is charged, no method dominates, and mitigation pays off only where it removes attenuation large enough to outweigh its cost.

5.1 Operating regimes

Table 3 shows 77 raw wins, 26 true practical ties, and distinct readout and ZNE subsets. Nine cells are inconclusive, eight because the 50-repeat completeness contract is not met. Raw wins require care: 40 win on accuracy and 37 on cost alone, and 63 have $|\mu_{\text{ref}}| < 0.06$, where noise moves the estimate toward a reference already near zero. Figure 1 shows that ZNE wins only at low and medium noise and readout only at medium and high noise.

Sensitivity to the practical threshold. The threshold 0.005 was fixed before the labels were computed, but it is a judgment, so Table 4 reruns the frozen label rule with other values. The number of raw wins depends strongly on it: at 0.01 there are 46 raw wins and 62 ties, and at 0.02 only 12 raw wins remain. Raw estimation mostly wins by small RMSE margins. Readout wins are stable (18 to 19), and ZNE wins shrink. On the signal set, raw's advantage near zero disappears entirely at 0.02 (0.0%), while ZNE's advantage at the boundary

Table 4: Main-grid labels and signal-set shares at four practical thresholds; the frozen value is in bold. Signal shares are at 4,096 shots.

Threshold	Raw	Tie	Read.	ZNE	Inc.	Near 0 raw	Edge ZNE
0.002	83	16	19	17	9	96.3%	66.7%
0.005	77	26	18	14	9	95.1%	63.5%
0.010	46	62	18	10	8	87.0%	57.9%
0.020	12	100	18	6	8	0.0%	51.6%

falls more slowly. The robust conclusion is therefore that no method dominates and that mitigated wins are concentrated in specific noise ranges, not that raw wins by a large margin. This analysis was added after all results were known.

Mitigation can also worsen estimation: 179 of 432 method summaries have negative RMSE gain relative to raw estimation. Readout produces 99 explicit hardware-efficient failures, concentrated at width 6, 1,024 shots, and medium/high noise. These failures are part of method performance, not missing data.

5.2 Signal strength and attenuation

The main grid confounds signal size with circuit family: every boundary-adjacent reference is GHZ and there is no non-GHZ strong signal. We therefore froze an independent set of 126 hardware-efficient circuits, selected by noiseless value alone into five bands of $|\mu_{\text{ref}}|$ from near-zero (below 0.06) to boundary-adjacent (at least 0.9), and ran them under seven noise conditions at 4,096 and 16,384 shots with 50 repeats.

Figure 2 shows the result. Near zero, raw is better in 95.1% of conditions at 4,096 shots; at the boundary, ZNE is better in 63.5%. The difference in ZNE's win share between these bands is 0.611 (interval 0.545 to 0.674). Raw's wins near zero are not accuracy: its median bias there is 43% of the reference. Within bands, ZNE's gain follows the raw attenuation it can remove (Spearman 0.55 to 0.92) more closely than the reference size (−0.12 to 0.39). By noise, the linear fit gains 0.020 under depolarizing and amplitude damping and 0.008 under dephasing, but loses 0.006 under coherent over-rotation, which inverse gates cancel and folding therefore cannot amplify. Global and local folding gave the same mean gain to three decimals.

5.3 Shots to reliability

Equation (4) predicts that a biased method cannot be rescued by shots. We tested this on two-candidate decisions sampled from exact channel probabilities, with budgets doubling from 1,024 to 2^{20} and 400 repeats. A target is reached at the smallest budget whose Wilson upper bound, and that of every larger budget, stays at or below it.

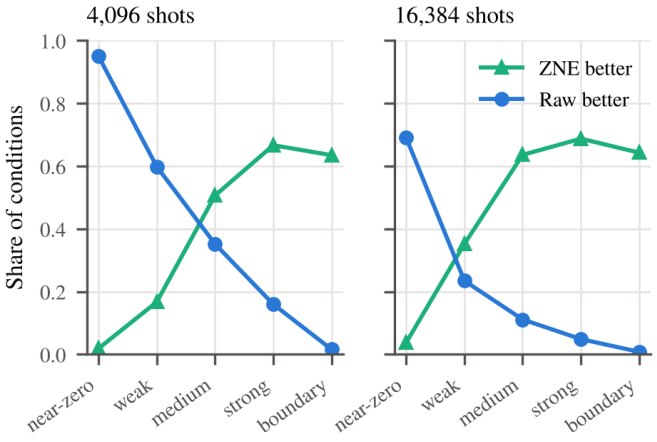


Figure 2: Share of noisy conditions where linear ZNE or raw is better by more than 0.005 in RMSE, by reference band, on 126 frozen circuits. Raw dominates only where the reference is near zero.

Across 40 positive-margin cells at a 10% target, raw reaches it in 26, readout in 33, ZNE in 27, and combined in 34; the rest are blocked by bias (13, 7, 11, and 5 cells) or not reached by 2^{20} shots. Figure 3 shows the three regimes. Without differential bias, raw or readout is cheapest in every cell, and ZNE or combined needs the same to eight times more shots. With differential readout bias, raw and ZNE error rises toward one while readout and combined converge. With differential gate-noise bias, only the folding methods help, and at high noise every method is blocked.

6 Estimator Quality Versus Decision Reliability

Takeaway: a method can lower RMSE and still choose worse, because the decision depends on the decisive gap, whose variance mitigation inflates.

6.1 Decision tasks

Expectation RMSE and decision reliability are distinct. The controlled combined-QEM decision study shows the separation directly. At margin 0.02, combined decision-error probability ranges from 0.26 to 0.48 across budgets and allocations and does not consistently beat readout alone. At margin 0.20, nearly all methods resolve the decision. Combined mitigation has the lowest expectation RMSE in 12 of 16 repository-precommitted conditions, yet all 16 of 16 combined-ZNE conditions fail at least one frozen validity criterion, and 12 of 24 paired factorial interactions are negative, so there is no uniform synergy.

6.2 MaxCut/QAOA candidate selection

For a graph $G = (V, E)$ and bitstring z, the cut value is $C_G(z) = \sum_{(u,v)\in E} \mathbf{1}[z_u \neq z_v]$. We use $p = 1$ QAOA [10]. Exhaustive enumeration fixes the optimum for one width-4 ring, one width-6 sparse graph, and one width-8 medium/dense graph, each with five frozen $p = 1$ QAOA candidates. Each method selects among all five within one budget $B \in \{4096, 16384\}$; the manifest contains 4,500 candidate-level rows and 900 selection groups.

All completed clear-ranking selections identify the ideal best candidate. Near ties separate estimator quality from decision quality. At 4,096 shots, near-tie error is 0.28 for raw, 0.32 for readout among completed selections, and 0.36 for ZNE; at 16,384 shots it is 0.17, 0.179, and 0.42. On the completed 4,096-shot ring and sparse strata, readout lowers energy RMSE from 0.183 to 0.148 and from 0.228 to 0.191, respectively, yet worsens top-1 selection on both; its medium/dense stratum fails entirely. Across completed repeats, the RMSE–selection-error correlation is only 0.153.

At 16,384 shots, mean regret is 0.0130 for raw, 0.0125 for completed readout selections, and 0.0378 for ZNE. Readout fails all 50 medium/dense selections at 4,096 shots and 22 of 50 at 16,384 shots because its assignment correction is ill-conditioned; these 72 failed groups are not treated as abstentions or successes. Results also vary by instance: ZNE's 16,384-shot top-1 error is 0.08 on the ring, 0.34 on the sparse graph, and 0.50 on the medium/dense graph. Certification is stricter still: only 2 of 828 completed selections certify a winner under simultaneous intervals. The remaining 826 completed groups are unresolved.

6.3 Mechanism: common mode and contrast

Figure 4 applies Eq. (3). In panel (a), every arrow from raw to ZNE points left (lower RMSE) and mostly up (worse selection). Panel (b) shows the medium/dense graph at both budgets. The common-mode term is small (an RMS of 0.09 against a raw RMSE of 0.81 at 16,384 shots), so energy RMSE is almost entirely contrast error, and ZNE lowers that contrast error in all six cells. But 77% of the MSE drop comes from one low-energy control candidate whose ideal energy (4.72) is far below the leader's (10.22). Selection depends instead on the gap between the top two candidates: ZNE shrinks its mean from 0.045 to 0.016 and raises its standard deviation from 0.053 to 0.137, and the normal approximation then predicts inversion rates of 0.20 and 0.45 against the observed 0.18 and 0.50. The share of the MSE held by non-contenders is specific to this graph (83–86% against 42–51% elsewhere), but the doubling of the gap's spread recurs in every cell (observed ratio 1.92 to 2.58). Off-diagonal error correlations across repeats are near zero for raw and ZNE, rejecting the hypothesis that raw rankings are stable because raw errors are shared.

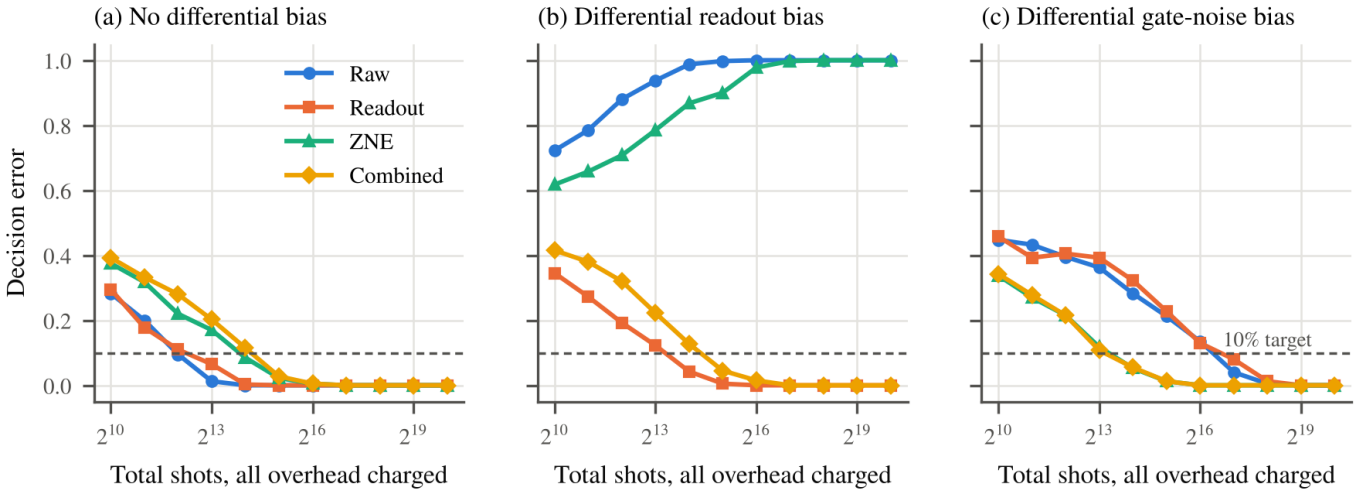


Figure 3: Decision error against total budget, all overhead charged, at margin 0.05 (a, b) and 0.10 (c), medium noise, 400 repeats. A method that keeps a differential bias gets worse with more shots.

Proposition 1 explains the inversion. Equation (5) requires ZNE to enlarge the mean contending gap by more than $\sqrt{4.375} \approx 2.09$ times. The ring and sparse graphs cannot reach that even with complete debiasing, because their raw gaps are attenuated only to about $a = 0.75$ ($\Delta/\Delta_r \leq 1.33$); the medium/dense graph could reach 3.73 ($a = 0.27$) but ZNE achieves a ratio of 0.36. The proposition also predicts that the sign of the preference does not change with budget, and the confirmatory comparison agrees: raw is better at both budgets, by a margin within the minimum difference at 4,096 shots and resolved at 16,384. The threshold was derived after these data were collected, so for new graphs it is a hypothesis.

Table 5: Consequence audit. Even without charging its diagnostics, the selector has more harmful outcomes than always-raw across all 180 evaluated cells.

Category	Sel.	Raw	Read.	ZNE
Correct, useful	63	106	19	18
Correct, tied	20	23	11	24
Incorrect, negligible	29	0	71	3
Incorrect, material	42	26	45	78
Catastrophic	17	16	25	48
Abstained, avoided failure	1	0	0	0
Abstained, unnecessary	5	0	0	0
Inconclusive	3	9	9	9

6.4 Confirmation on new repeats

The mechanism analysis was exploratory, so we froze a confirmatory plan before collecting new data. From 34 committed comparisons, a power analysis on the observed discordant-pair rate set the repeats each needed for 80% power against a minimum difference of 0.10 in decision error or 0.15 in selection error, with Holm correction across the family. Null results are labeled *close within margin* when the interval lies inside the minimum difference, and *underpowered* otherwise.

The core counterexample holds: on the medium/dense graph at 16,384 shots, raw errs in 0.271 of selections and ZNE in 0.379 over 398 new paired repeats ($p = 0.001$). At 4,096 shots the two are within the margin. Overall, 13 comparisons resolve a difference, 12 are close within the margin, and 9 remain underpowered even at their planned repeats, so the v1-based power estimates were optimistic for those. On 72 comparisons from two task settings the original grid never ran, 35 resolve, 34 are close, and 3 are underpowered.

6.5 Selecting a method adaptively

Frozen rule selector. A non-ML rule selector uses width, depth, declared noise regime, shots, calibration conditioning, and ZNE fit residuals. On 29 held-out cells it reaches 41.4% decision compatibility. Its diagnostics are 50-repeat cell means that a runtime policy would not have, so this is an optimistic baseline. Table 5 shows what its choices cost.

Across all cells, the selector has 59 material or catastrophic outcomes and always-raw has 42. All of raw's harmful outcomes occur in the 84 cells with $|\mu_{\text{ref}}| \geq 0.06$, so raw is not a general policy either.

Among diagnostics, condition number acts only as a warning: readout is reliable in 75% of cells below 1.5, 50% from 1.5 to 3, 10% from 3 to 8, and none of the 4 cells at 8 or above. ZNE fit residual does not discriminate, with rates of 24–33% in every adequately sampled bin. Three cells flip across the condition-number gate of 8 under repeat resampling, with means of 7.87, 8.10, and 8.16. Table 6 lists every single-feature level supported

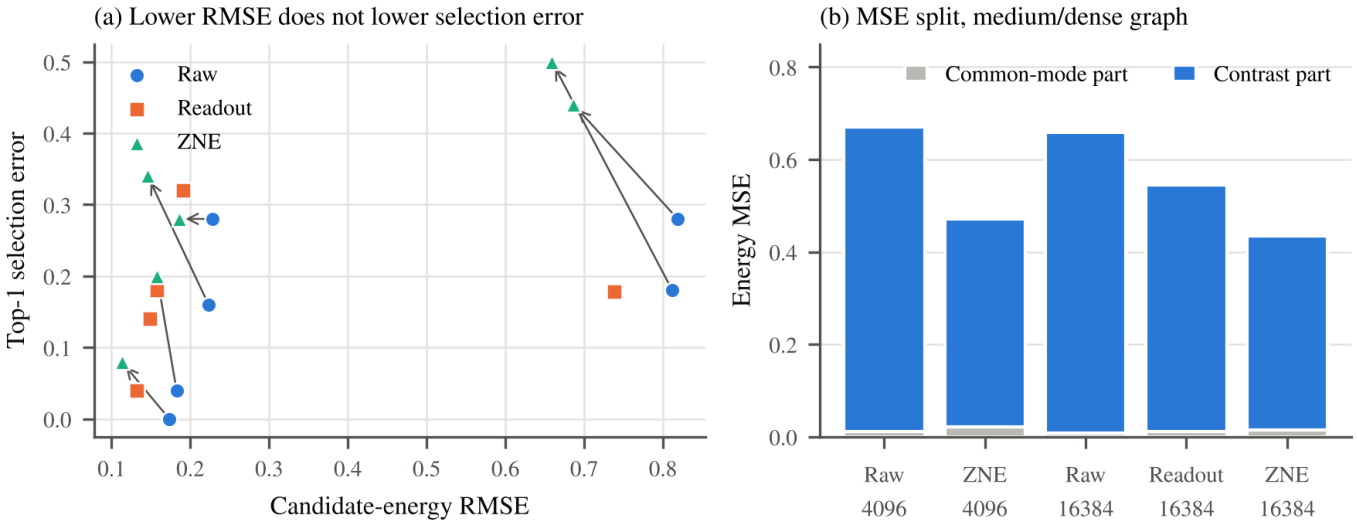


Figure 4: (a) Energy RMSE against top-1 selection error for each QAOA graph, budget, and method; arrows go from raw to ZNE in the same cell. (b) Energy MSE split into the common-mode and contrast terms of Eq. (3) on the medium/dense graph.

Table 6: Single-feature levels supported for reliable use. None exists for ZNE.

Method	Level	Reliable	Stable
Raw	width 6	36/44	55%
Raw	noise zero	32/32	100%
Raw	$\|\text{ref}\| < 0.06$	56/56	100%
Raw	margin 0.2	11/12	93%
Readout	noise zero	29/32	89%
Readout	$\|\text{ref}\| \geq 0.5$	26/30	88%
Readout	margin 0.2	11/12	94%

Table 7: Held-out selector comparison, 20 tasks and 14,000 runs. Reliable coverage is the share of runs that are covered and correct. The oracle picks the best baseline per condition after scoring.

Policy	Coverage	Regret	Reliable	Gate shots
Always raw	1.00	0.0024	0.919	113,664
Always readout	1.00	0.0031	0.908	90,931
Always ZNE	1.00	0.0059	0.867	340,992
Always combined	1.00	0.0072	0.855	272,794
Selector (v3)	0.64	0.0031	0.630	147,767
Abstaining readout	0.68	0.0033	0.666	147,763
Oracle baseline	1.00	0.0022	–	–

for reliable use; none exists for ZNE, and the frozen joint envelope is empty.

Repository-precommitted selector test. A runtime selector sees only one repeat's pilot, which spends 25% of the budget. The selector test uses a fresh grid committed before it ran, with 120 cells and 24,000 rows. All three frozen criteria fail, so the runtime selector is a no-go. For H1, 14 of 26 decided cells are reliable (lower bound 35.5%). For H2, 27 of 94 no-safe cells avoid a failure (lower bound 20.6%). For H3, mean regret on decided cells is 0.0059, against 0.0606 for always-raw and 0.0033 for always-readout. As an exploratory reading, raw is reliable in all 36 cells with $|\mu_{\text{ref}}| < 0.06$ and readout in 34 of 48 with $|\mu_{\text{ref}}| \geq 0.5$, but a runtime policy does not know the reference.

Selector with abstention (v3). The third selector chooses the largest of three candidate values. It spends 10% of the budget on calibration and 20% on a pilot at folds 1, 3, 5, defaults to readout correction, switches to combined mitigation only when both pilot contenders show a resolved rise in magnitude after extrapolation, may fall back to raw when correction is unresolved on every candidate, and abstains when its two calibrations disagree or the final top gap is unresolved. The rule was chosen from a 30-rule grid on training and validation tasks drawn from four circuit families (hardware-efficient, rotated GHZ, transverse-field Ising Trotter steps, and ring QAOA; 72 tasks in total) and scored once on 20 held-out tasks. Table 7 shows the result: forced regret 0.0031 against 0.0031 for always-readout, at 1.63 times the two-qubit gate executions, and lower reliable coverage than the same abstention rule applied to readout alone. No frozen acceptance route passed, so adaptive selection did not beat a simple fixed strategy. Always-raw had the lowest regret in every holdout group: the candidates share one readout, so its bias is common-mode and cancels in the comparison.

7 Validity and Operational Boundaries

Takeaway: the runtime checks we tested see sampling error, drift in marginals, and resolved decays, but not differential bias or a change in correlation structure.

7.1 ZNE validity gates

An extrapolate is not credible merely because it moves toward the reference. The first gate requires retained signal, physical bounds, positive gain with uncertainty, an advantage over matched-cost garbage folding, and few destroyed or unstable repeats. It accepts only 5 of 16 simulated cells, all Bell-state conditions, and none of the 8 hardware-efficient cells, matching concerns about artefactual ZNE improvement [16]. A post hoc audit found two weaknesses: two of its checks read the exact reference, and its trend check partly double counts physical-bound violations.

We therefore chose a runtime-only gate from 256 candidates (four physical-bound variants, eight trend variants, and optional garbage, retention, and estimated-gain checks) on training and validation circuits of the frozen signal set, and scored it once on 30 held-out circuits. "Really improved" means the fit's RMSE, from independent draws, beats raw by more than 0.005. The frozen choice reduced to a single slope-sign stability check. Figure 5 shows the holdout: false acceptance 0.176 and false rejection 0.000, against 0.520 and 0.284 for the runtime checks of the first gate and 0.018 and 0.454 for the full first gate, which reads the exact value. Every one of the 230 holdout rejections names its failure mechanism. The selected check measures whether the decay across folds is resolved, not whether the extrapolate is right, so its false acceptance rises with shots (0.131 at 4,096, 0.224 at 16,384) and reaches 0.414 under amplitude damping. Leave-one-factor-out, weighted-residual, and quadratic-agreement checks scored below chance, and the model selection rule (local folding with an exponential fit) did not beat the default linear fit on holdout.

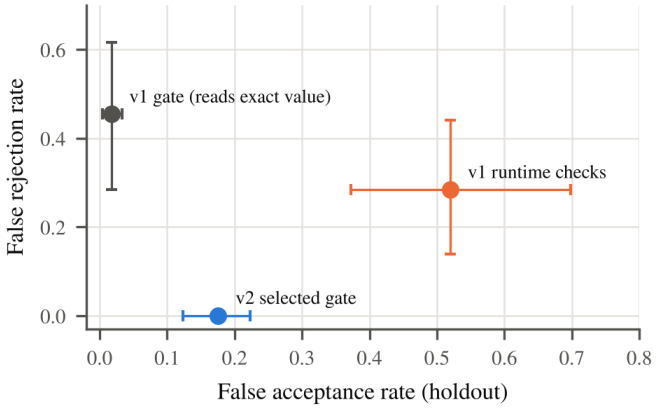


Figure 5: Holdout error rates of three ZNE gates on 30 circuits not used to choose them. Bars are circuit-cluster bootstrap intervals; the v2 gate had no false rejection in any resample.

7.2 Certification

Sequential stopping. Two candidates are sampled in batches until an interval for their gap excludes zero [30]; every look is tested at α/K, a conservative alternative to confidence sequences [13]. Corrected looks work as intended against sampling error, keeping false certification below 3% at zero margin, compared with 22–25% for uncorrected peeking. At margin 0, where any certification is false, Wilson stopping certifies in 1.7%–2.7% of repeats, whereas uncorrected peeking reaches 22.4%–24.9%. Table 8 shows the trade-off: stopping early saves shots but leaves more decisions unresolved. Certification does not detect bias: with strong gate-noise imbalance at margin 0.05, raw certifies the wrong candidate in 100% of repeats after a mean of 2,022 shots, that is, after about 2,000 shots, and ZNE still does so in 77%.

Table 8: Stopping early saves shots but leaves more symmetric-task decisions unresolved (16,384 shots, margins above 0, nonzero noise).

Method	Policy	Shots	Unres.	Wrong
Raw	Fixed budget	16,384	27.1%	0.08%
Raw	Wilson	9,583	40.7%	0.00%
Raw	Gap confidence	10,653	48.2%	0.00%
Raw	Variance aware	14,136	70.8%	0.00%
Raw	Naive peeking	5,067	16.2%	3.25%
Readout	Wilson	11,429	43.7%	0.08%
ZNE	Wilson	13,045	63.6%	0.08%

Two-layer certification. On the 20 held-out selector tasks, five independent batches each carry their own calibration. Layer 1 requires the winner to beat every challenger by a one-sided t test on batch differences, which keeps shared calibration within a batch rather than assuming independence. Layer 2 requires six bias alarms to stay silent: raw and corrected winners differing, a reversal of the gap at fold 5, calibration drift between batches, an out-of-bound value, variance amplification, and batch sign instability. Both layers certify 0.502 of runs with false certification 0.0074 per run; Layer 1 alone certifies 0.518; a naive test that treats all estimates as independent certifies 0.700 with 0.0175 false certifications per run. Figure 6 shows where the errors are. The alarms fired in at most 2.4% of Layer-1 passes, so the second layer barely changed anything. Near-tie tasks fail: 35.9% of certified runs pick the wrong winner, because the bias survives every layer. Coherent over-rotation has the highest false certification of any noise, 6.9% of certified runs.

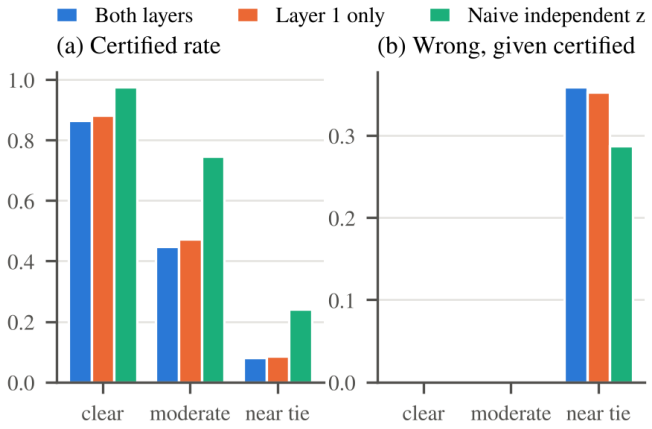


Figure 6: Certification on 20 held-out tasks by ideal gap class. Correcting for shared samples (blue, orange) lowers false certification on clear and moderate tasks, but no rule controls it on near ties.

7.3 Readout correction at scale

Under one shared budget, full-matrix correction gives each of 2^w calibration states very few shots: 3 at width 8 and 4,096 shots. It is ill-conditioned or singular in 93.0% to 100.0% of width-8 runs at 4,096 shots and fails always at width 10 (Fig. 7). At 16,384 shots it completes at width 8 but is worse than no correction. Given 1,024 shots per calibration circuit instead, it completes at widths 8 and 10 using 80.0% to 98.5% of all shots for calibration. Full-matrix failure is therefore a resource failure, not a method failure. The model matters more than the budget: under pair-correlated readout, tensor-product, observed-subspace, and matrix-free correction all assume independent errors and become worse than no correction, while pairwise-local correction stays below 0.05 at every width. Observed-subspace and matrix-free solvers match tensor-product correction wherever it is correct.

7.4 Calibration mismatch and early warning

We calibrated once on A_{cal} and measured under $A_{\text{eval}}(\lambda) = (1-\lambda)A_{\text{cal}} + \lambda A_{\text{mis}}$ for 21 values of λ. Mitigation lost to raw only under asymmetric drift (from $\lambda = 0.25$ to 0.60, later at larger width) and under a change from independent to fully pair-correlated flips with equal marginals (from 0.50 to 0.75; Fig. 8). Under abrupt change, calibration age, and added pair flips it stayed better than raw up to $\lambda = 1$, because raw suffers the same errors. As predicted, the calibration condition number never warned (0.0% of cases), since it comes from the stale calibration. A 1,024-shot check of the two basis states warned in time in 76.2% of cases, but it is blind to the change to correlated flips, because the marginals it measures do not change, and that is one of the two scenarios where mitigation loses. Decision error never rose steeply at margin 0.10, so that question is unanswered by this design.

Table 9: A triggered policy approaches every-decision reliability at lower cost (first drift study, 100 decisions).

Recalibration	Shots per decision	Error
Every decision	2,048	0.062
Every 10 decisions	1,126	0.074
Uncertainty trigger	1,229	0.067
Never	1,034	0.234

Table 10: Frozen simulator predictions for proposed device cells at 16,384 shots, or a 100-decision workload for calibration reuse. These are hypotheses for a device, not hardware results. Val. is single-method ZNE validity; Mshots is the planned total in millions.

Cell	Raw	RO	ZNE	Comb.	Val.	Mshots
Readout-dominated	0.740	0.893	0.791	0.958	credible	3.3
ZNE control	0.970	0.987	0.980	0.997	credible	4.1
ZNE signal risk	0.232	0.263	0.265	0.305	none	4.1
Near tie	0.012	0.021	0.014	0.021	none	41.0
Clear margin	0.179	0.181	0.184	0.188	none	3.3
Calibration reuse	0.510	0.530	–	–	–	16.3

7.5 Calibration reuse under drift

Table 9 compares policies over 100 decisions in the first drift study. Triggered recalibration has decision error close to recalibrating every time at about 40% lower cost, while never recalibrating fails under directed drift. A larger sweep over drift amplitude, timescale, shape, run length, margin, calibration cost, and trigger threshold, with all check shots charged, gives the same picture (Fig. 9). A trigger at $z = 2$ matches or beats every-decision recalibration in mean decision error with about 60% of the shots, except under sinusoidal drift at amplitude 0.06; stricter triggers lose there. Every-decision recalibration also adds fresh calibration noise at each step.

7.6 Hardware readiness

Table 10 lists frozen predictions for six candidate device cells. The review decision is no-go: only 1 of 7 go requirements is fully met, and all 6 no-go triggers apply. The plan would need about 72.1 million shots. The positive control is credible at 16,384 shots but rejected at 4,096, three of five ZNE-relevant cells have no single-method validity assessment, and no code captures device metadata. This is a measured outcome of the review, not a statement about how any device would behave, since no job was submitted.

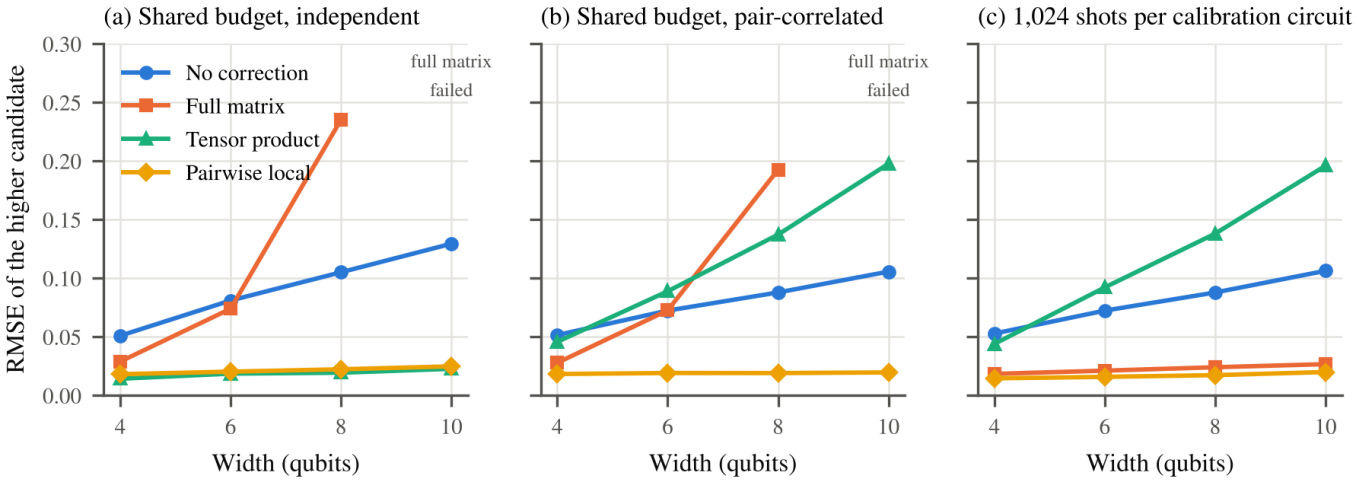


Figure 7: RMSE of a parity observable on qubits 0 to $w-2$ against width, 200 repeats. Under a shared budget, full-matrix correction fails by width 8 to 10; with correlated readout, an independent model does worse than no correction.

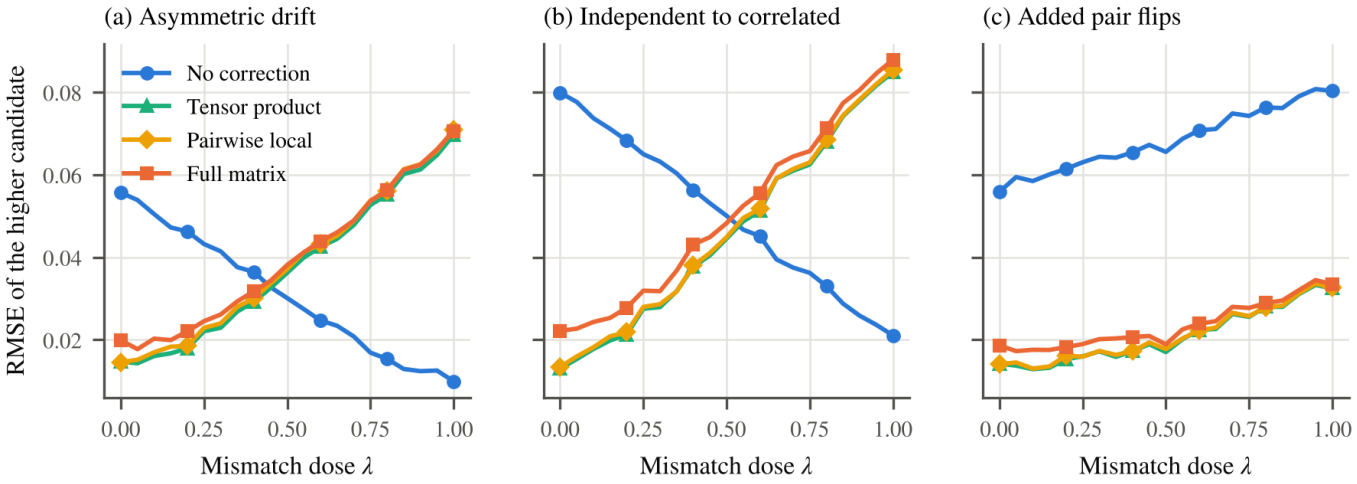


Figure 8: Readout mismatch dose-response at width 6, 150 repeats. Corrections lose to raw under asymmetric drift (a) and a change to correlated flips (b), but not under added pair flips (c).

8 Negative Results by Mechanism

Table 11 classifies each negative or inconclusive result. A *method* failure persists with unlimited shots; a *resource* failure disappears with more calibration or target shots; a *model* failure comes from a wrong noise or readout model.

9 Verification and Reproducibility

Raw repeat-level rows are never replaced by summaries. `src.verify_final_project` recomputes the main summaries from raw rows at tolerance 10^{-9}, accounts for every manifest, and checks figures and this document. `src.claim_ledger` separately checks 23 computed claims covering 195 printed numbers without importing project code, and labels computed results, manual interpretation, hypotheses, limitations, and review judgments. Of these claims, 11 are recomputed from raw repeat rows, 9 from processed tables (for example, the signal study's truth table, whose fit errors agree with normal theory to within 0.0014), 2 from manifests, and 1 from a processed table only. Every number generated for Sections 5 to 7 is recomputed except one ratio of gate executions, which needs the project's circuit builders. Five deterministic sampled claims are recomputed from repeat rows after a correction to the audit implementation. Nine generated tables and the figures have source provenance but have not all been independently recomputed, so the paper as a whole is not independently verified. Replacing the original BCa interval with a deterministic percentile-bootstrap sensitivity calculation did not change the 144 cell labels.

Table 11: Negative and inconclusive results. "Driven by" is deterministic (D) or sampling (S); "More shots" says whether a larger budget removes the failure; "Runtime signal" names a diagnostic that predicted it, if any. These are review judgments based on the reported data.

Result	Driven by	Type	More shots	Runtime signal	Reproduced across
Raw beats mitigation in most main-grid cells	D	Resource	No (cost is structural)	Small $\lvert\hat{\mu}\rvert$	Grid, signal set, selector tasks
Lower RMSE, worse QAOA selection	D + S	Method	No (variance factor 4.375)	None found	3 graphs; confirmed at 16,384
Biased method misses a decision target	D	Method	No; error rises	None	3 task types
ZNE gate false acceptance	D	Method	No; rises with shots	Slope check is the gate	Holdout, all noises
Model-selection rule underperforms	S	Statistical	Possibly	–	One holdout
Adaptive selectors (v1, v2, v3)	D + S	Resource	Unknown	–	Three designs
Near-tie false certification	D	Method	No	None	All noises
Bias alarms rarely fire	D	Method	No	–	One holdout
Full matrix fails at width 8	D	Resource	Yes, at 80–98.5% calibration share	Condition number	Widths 8, 10; 3 models
Independent model under correlated readout	D	Model	No	Pairwise calibration	Widths 4–10
Check circuit blind to correlation change	D	Model	No	–	Widths 4–8
No steep decision-error rise found	–	Design	–	–	Inconclusive

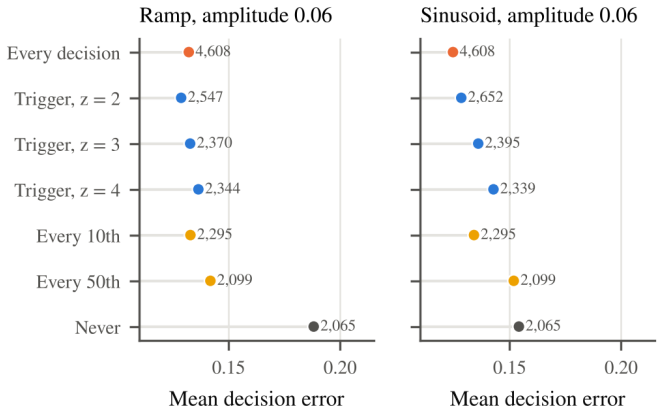


Figure 9: Recalibration policies under drift of amplitude 0.06, averaged over the other sweep parameters. Labels give total shots per decision, including calibration and check circuits.

An artifact registry declares every generator, its configs, manifests, outputs, and expected rows, and `src.artifact_provenance` records a content hash for every tracked artifact. Continuous integration rebuilds every routine generator from a clean clone on each push. The 16 long generators of the release scope are rebuilt by a separate workflow, one job per generator, each deleting that generator's committed outputs in a clean clone, running it with the locked dependencies, and comparing the result with the committed files, which were generated on Windows. Two runs on Linux reproduced 12 of the 16 generators within tolerance, including those behind the confirmatory, QAOA, decision, combined-QEM, shots-to-reliability, mismatch, and drift results. One of these failed in the first run only because the registry assigned a figure to the wrong generator, which is now fixed. Four do not match the Windows files. In three the difference changes no reported result: the selector-v3 estimates differ only in the sign of a zero, the readout-feasibility condition number is infinite on one platform where it is large and finite on the other, and the signal study's out-of-bounds rate, which no analysis reads, differs in 9 of 17,640 rows. The fourth is substantive: in 29 of 5,760 scalable-readout runs, whether an ill-conditioned correction completes depends on the platform, which changes 5 of 192 summary cells. That study, of which this paper prints only the planned repeat count, should be read as platform dependent near its conditioning limit. The committed results remain the Windows results and no tolerance was changed. The full record is in `docs/release_rebuild_v1.md`.

`src.independent_reference` reproduces exact values within 10^{-12} and, from fixed counts, all completed QAOA selections. It does not check noisy sampling. Width-8 readout energies differ by up to about 10^{-6}. The later studies use a density-matrix simulator checked against Aer. Both implementations were written by the same author.

10 Discussion

What the evidence supports. The strongest conclusions are structural rather than empirical accidents. Equal accounting reverses rankings whenever no decision-relevant bias exists, because calibration and folding cost shots that raw estimation spends on the target. Bias cannot be bought off with shots, and every interval-based tool we tested, from sequential stopping to two-layer certification, controls only sampling error. Runtime diagnostics see changes in marginals and resolved decays, not hidden bias or a change in correlation structure. And a

correctly modeled local correction succeeds where a mismodeled one does worse than nothing.

Relation to the limits of mitigation. The exponential bounds of [26, 27, 24] say that mitigation becomes unaffordable as circuits grow. Our results add that it can fail to pay off well before that point: at widths 2 to 8, where its cost is modest, the decision still worsens whenever Eq. (5) fails. A mitigation method should therefore be judged against the decision it serves, not only against the error of the value it returns.

Adaptive analysis across the project. Each study froze its own rules before its run, but the sequence of studies was chosen after seeing earlier results: the signal study followed the main grid's confounding, the confirmatory study followed the QAOA inversion, and the readout-boundary study followed earlier readout failures. At the level of the whole project the analysis is therefore adaptive. Only the confirmatory comparisons were planned with a stated power and family-wise error rate; the other findings should be read as exploratory or as tests of one frozen study at a time, and the many studies increase the chance that some reported contrast is a false positive.

Rule freezing and independence. The combined, QAOA, scalable-readout, runtime-selector, signal, gate, confirmatory, selector-v3, and readout-boundary protocols were committed before their result artifacts. The first ZNE gate's checks were named in advance, but its numerical thresholds and results first appeared together in the same repository state; therefore this is not an independently timestamped preregistration. The sequential and calibration studies and the first selector use frozen repository rules that first appear with their results. Commit order proves repository publication order only, not the wall-clock order of design or execution. A registration with an independent timestamp, such as those discussed by Nosek et al. [22], would remove this limitation in future work. In the readout-boundary study, a 20-repeat smoke run showed that an all-qubit parity is blind to pair flips; the observable was changed before freezing, and no hypothesis, threshold, or grid value changed.

Limitations. All noisy values are simulator-derived with synthetic noise; there is no device drift, calibrated crosstalk, leakage, or non-Markovian noise. The main grid has one hardware-efficient circuit per width and depth, so width and depth effects are confounded with specific angles. The QAOA study has three curated graphs with $p = 1$; its candidates are frozen inputs whose exact-tie ordering depends on the generating machine. The signal and selector holdouts have 30 circuits and 20 tasks, and candidates in the selector tasks share one readout, which favors raw estimation. The scalable-readout manifest planned 960 repeats of correlated readout at widths 8–12 that were never run; correlated models ran only at widths 2–6 in the original mismatch study and in the separate readout-boundary protocol, whose rows are not merged into the original table. The drift sweep uses one qubit. The practical threshold changes the count of raw wins substantially (Table 4). Only four method families are compared; probabilistic error cancellation, Clifford data regression, and randomized compiling, which converts coherent errors such as the over-rotation that defeated both the gate and certification into stochastic ones [31], are not. The independent reference implementation was written by the same author; an implementation from another code base, such as Mitiq's ZNE [17], would be a stronger check.

Why the hardware review ends in no-go. A disagreement between a device and Table 10 would first falsify the transfer of a synthetic noise model, not the simulated conclusion. Separating drift from method failure needs interleaved circuits, repeated reference controls, and a calibration record around each block, none of which exists here. A run now would mainly produce a demonstration, so a no-go is the more honest result. The cheapest informative device question is whether real readout carries enough pair correlation to make independent models worse than none, which needs calibration circuits only.

11 Conclusion

No QEM method is uniformly useful under fixed-budget accounting, and better estimation is not better decision-making. Mitigation pays off when it removes a bias that differs between the alternatives being compared and that is larger than its own variance cost, which Proposition 1 makes precise as $\Delta_{\mathrm{m}} > \sqrt{\gamma}\,\Delta_{\mathrm{r}}$; it does not pay off when the error is common-mode, when the method's model is wrong, or when its calibration consumes the budget. Adaptive selection, statistical certification, and the diagnostics we tested could not identify these cases reliably at runtime, so simple fixed policies remain the reference that any adaptive rule must beat, and bias detection is the central open problem.

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A Reproducibility

```
python -m pytest
python -m src.verify_generated_artifacts
python -m src.verify_generated_artifacts \
    --scope release --module src.generate_signal_v2
```

Table 12: Evidence matrix for the original tracks. Y marks evidence of that type; An. is analytic, NL noiseless simulation, NS noisy simulation, HW hardware.

Track	An.	NL	NS	HW	Supported finding	Unresolved question	Limitation
Expectation accuracy	Y	Y	Y	–	No universal winner: 77 raw, 26 ties, 18 readout, 14 ZNE.	Which method wins when the signal is strong and the circuit is new?	Raw wins are mostly near-zero references; one circuit per width and depth.
Decision reliability	Y	Y	Y	–	Lower RMSE does not ensure a better decision; near-tie error is 0.26-0.48 for combined QEM.	How does this extend beyond two candidates?	Controlled margins on small circuits.
ZNE validity	Y	Y	Y	–	Apparent gain is not validated gain: 5 of 16 cells pass, none hardware-efficient.	Does the gate hold for other circuits and fits?	Two circuits, linear fit, and folds 1, 3, and 5.
Combined mitigation	–	Y	Y	–	Lowest RMSE in 12 of 16 conditions, yet all 16 validity checks fail.	Are readout and gate errors separable on a device?	Six cells, two budgets, two allocations, higher overhead.
Readout scalability	Y	Y	Y	–	Lower memory is not robustness: mismatch error reaches 0.88-0.94.	How large is real correlated readout error?	Synthetic channels; correction amplifies sampling spread.
Application case study	Y	Y	Y	–	Lower energy RMSE does not imply better selection; 2 of 828 completed selections certify.	Does this hold for other graphs and depths?	Three graphs, five frozen candidates, and p=1.
Calibration drift	–	–	Y	–	Triggered recalibration saves about 40% of shots with similar error; never recalibrating reaches 0.386 under abrupt drift.	What drift does a real device show?	Directed synthetic stress; not a physical lifetime.
Sequential certification	Y	Y	Y	–	Corrected looks keep false certification at or below 2.7%, versus up to 24.9% for peeking; bias is not detected.	How can bias be detected before certifying?	Two-qubit circuits, one observable, conservative bounds.
Hardware readiness	–	–	Y	–	Review decision: no-go. No job was submitted.	Can a backend-specific, powered, controlled protocol be justified?	Predictions depend on simulator noise; no backend data.

```
python -m src.generate_signal_v2 --workers 2
python -m src.zne_gate_v2
python -m src.confirmatory_v1
python -m src.confirmatory_qaoa_v1
python -m src.shots_to_reliability
python -m src.generate_selector_v3 --workers 4
python -m src.generate_readout_feasibility_v2 --workers 4
python -m src.generate_readout_mismatch_dose --workers 4
python -m src.generate_drift_policy --workers 4
python -m src.generate_paper_v2_assets
python -m src.artifact_provenance --write
python -m src.verify_final_project
python -m src.claim_ledger
```

The full release scope runs as one job per generator in `.github/workflows/rebuild-generators.yml`. Protocols are in `docs/`: `signal_v2_protocol.md`, `parts_7_8_protocol.md`, and `part_9_protocol.md`, with results in `signal_v2_results.md` and `parts_7_9_results.md`.

B Gate definitions

The first ZNE gate passes a cell when the lower 95% bound of final-fold retention exceeds 0.10, the physical overshoot is at most 10^{-12}, signed-probability negative mass is at most 0.05, the lower bounds of the garbage-control advantage and of the paired gain exceed zero, no repeat is signal destroyed, and at most 5% are unstable. The second gate's candidates and selection rule, the selector grid, and the certification alarms are listed in the protocol files above, with every threshold fixed before the corresponding run.

C Evidence by track

Table 12 separates each original finding by evidence type. The hardware column is empty for every track.